

Are the β_0 transverse logarithms in NLO gluon production running-coupling logs, or DGLAP?

Transposing Kovner–Lublinsky–Skokov–Zhao (arXiv:2308.15545) to single-inclusive gluon production at $O(g^4)$

Every algebraic claim marked **VERIFIED** was checked symbolically (SymPy) or numerically; the script is `verify.py` in this directory. Conventions are those of the two $\rho\rho$ expressions supplied by the user; where an overall sign depends on an index-ordering convention not fixed by those expressions, this is said explicitly.

0. Summary

Yes — the effect is present, and the two $\rho\rho$ terms are exactly the right place to look. They are in one-to-one correspondence with the two β_0 -carrying kernels of the NLO JIMWLK Hamiltonian:

- the $\mathfrak{A}^{\dagger(1)} \otimes \mathfrak{A}^{(3)}$ term $\leftrightarrow K_{JSJ}$ (one Wilson line), where the $\ln \mu^2$ coefficient is *twice* what the running coupling can absorb;
- the $\mathfrak{A}^{\dagger(1)} \otimes \mathfrak{C}_1^\dagger UU \mathfrak{B}_2$ term $\leftrightarrow K_{JSSJ}$ (two Wilson lines), whose $Z \rightarrow 0$ divergence supplies the other half — and that half is DGLAP, not the β -function.

The identification is not an analogy. Three things are exact:

1. $\mathfrak{C}_1$ is the light-cone $g \rightarrow gg$ splitting vertex: its square is $P_{gg}(\zeta)/2N_c$, identically KLSZ's $[\xi^2 + (1 - \xi)^2 + \xi^2(1 - \xi)^2]/\xi(1 - \xi)$.
2. The $\delta^{(2)}$ in $\mathfrak{C}_1$ puts the parent at the momentum-fraction-weighted *centroid* of the daughters — identically KLSZ's coordinate-space splitting kinematics.
3. $\mathfrak{B}_2$, in the merging limit, factorizes *exactly* into $\mathfrak{A}^{(1)} \otimes \mathfrak{C}_1$, prefactors and all.

Consequently the coefficient of the transverse log is forced to be KLSZ's $\beta_0^g = 11N_c/3$ — there is no freedom left.

Two results are new to the production case and do not appear in KLSZ:

- Because the observed gluon sits at the *centroid* $w = (1 - \zeta)z + \zeta x$, the measurement phase $e^{-ik(w'-w)}$ carries **no** dependence on the pair separation Z . So k_T does **not** regulate the Z integral. It enters only through the emitter–emission distance $X = y - w$, which the Fourier transform fixes at $|X| \sim 1/k_T$.
- The consequence is the opposite of what one might guess: the residual DGLAP-like logarithm is $\ln(Q_s^2/k_T^2)$, and it is large in the **saturation region** $k_T \lesssim Q_s$, not at large k_T . The resummed correction is a power $(Q_s^2/k_T^2)^{\alpha_s \beta_0^g/4\pi}$ multiplying the Wilson line — a factor of order 2–4 for $k_T \sim Q_s/3$ at realistic α_s (Sec. 9).

Correction to an earlier estimate. In discussion before this note I suggested that the phase would cut the Z integral at $|Z| \sim 1/(\zeta k_T)$, making the DGLAP log $\ln(k_T^2/Q_s^2)$ and the effect large at *large* k_T . Working in centroid variables (Sec. 8) shows that is wrong: the Jacobian $\partial(w, Z)/\partial(z, x) = 1$ and the phase depends on w alone. The log is $\ln(Q_s^2/k_T^2)$ and the effect lives at *small* k_T . The rest of the earlier identification is unchanged.

1. Notation

Throughout, $\zeta \equiv p^+/k^+ \in (0, 1)$ is the momentum fraction of the daughter gluon at transverse position x ; the other daughter, at z , carries $k^+ - p^+$. Transverse vectors are two-dimensional and Latin indices i, j, k, m, n are transverse polarizations. U^{ab} is the adjoint Wilson line of the target field; ρ^a is the projectile colour source. Following the supplied expressions,

$$\mathfrak{A}_i^a(k^+, \mathbf{z}) = \frac{ig}{\sqrt{2}\pi\sqrt{k^+}} \int_{\mathbf{x}} \frac{(\mathbf{x} - \mathbf{z})^i}{(\mathbf{x} - \mathbf{z})^2} \rho^a(\mathbf{x}),$$

$$\mathfrak{C}_{1jki}^{bca}(k^+, z; p^+, x) = -\frac{gf^{abc}}{2\pi\sqrt{2k^+}} \frac{\sqrt{p^+(k^+ - p^+)}}{k^+} \delta^{(2)}\left(z + \frac{p^+}{k^+}x\right) \frac{x^m}{x^2} \left[\delta_{im}\delta_{jk} - \frac{k^+}{p^+}\delta_{jm}\delta_{ik} - \frac{k^+}{k^+ - p^+}\delta_{km}\delta_{ij} \right],$$

and $\mathfrak{B}_2$ as supplied. In $\mathfrak{C}_1$ the index i is the parent polarization, j the daughter carrying p^+ , k the daughter carrying $k^+ - p^+$; this assignment is forced by which $1/p^+$ or $1/(k^+ - p^+)$ multiplies which δ , and it is the assignment under which the merging lemma of Sec. 4 holds.

The leading-order cross section, which everything below is measured against, is the $O(g^2)$ $\rho\rho$ term with the same outer structure:

$$\frac{d^3 N_{\text{LO}}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{w, w'} e^{-ik(w'-w)} \left[-\bar{\mathfrak{A}}_i^{\dagger(1)a}(w') + \mathfrak{A}_i^{\dagger(1)b'}(w') U^{b'a}(w') \right] \left[U^{ab}(w) \mathfrak{A}_i^{(1)b}(w) - \bar{\mathfrak{A}}_i^{(1)a}(w) \right].$$

Its kernel is the product of two Weizsäcker–Williams fields, $\frac{X^i Y^i}{X^2 Y^2} = \frac{X \cdot Y}{X^2 Y^2}$, with

$$X \equiv y - w \quad (\text{amplitude}), \quad Y \equiv y' - w' \quad (\text{conjugate}),$$

y, y' being the projectile source positions. This is literally the LO JIMWLK kernel of KLSZ Eq. (2), so their X, Y and ours are the same objects and their $\bar{Q}^2, \tilde{Q}^2$ translate without modification. The one structural difference — and it matters in Sec. 8 — is that in the Hamiltonian the gluon is emitted at the *same* point z in amplitude and conjugate, whereas here $w \neq w'$ because we Fourier transform at fixed k .

2. The dictionary

NLO JIMWLK (KLSZ Eq. 5)	Production at $O(g^4)$, two ρ	Content
$\int_{x,y,z} K_{JSJ} [J_L J_L + J_R J_R - 2J_L S J_R]$	$\mathfrak{A}^{\dagger(1)} \otimes \mathfrak{A}^{(3)}$ (second expression)	one Wilson line; carries $\ln \mu_{\text{MS}}^2$ with coefficient $2\beta_0$
$\int K_{JSSJ} [J_L D(z, z') J_R - \frac{N_c}{2} (J_L J_L + J_R J_R)]$	$\mathfrak{A}^{\dagger(1)} \otimes \bar{\mathfrak{C}}_1^\dagger U(x) U(z) \mathfrak{B}_2$ (first expression)	two Wilson lines; UV divergent as $x \rightarrow z$
$D^{ad}(z, z') = \text{Tr}[T^a S(z) T^d S^\dagger(z')]$	$U^{db}(x) U^{ce}(z)$ contracted with f^{dca}	the resolved pair
$\int_z \frac{X \cdot Y}{X^2 Y^2}$	$\mathfrak{A}^{\dagger(1)} \otimes \mathfrak{A}^{(1)}$	LO, Lipatov vertex squared
$J_L^a(x) J_L^b(y), J_L^a(x) S^{ab}(z) J_R^b(y)$	$\rho^a(y) \rho^b(y')$ bilinears	dilute-projectile limit — the "two ρ " restriction
$K_{q\bar{q}}$	<i>absent</i> from the supplied terms	needed for $\beta_0^q = -2N_f/3$ (Sec. 10)

The restriction to two powers of ρ is exactly the dilute-projectile limit in which the JIMWLK rotation operators reduce to $J \sim \rho$, so the supplied terms are precisely the sector of the production cross section in which KLSZ's analysis lives: one emitter in the amplitude, one in the conjugate.

One point of interpretation is worth making at once, because it decides which DGLAP evolution is at stake. In the first expression the ordering is, reading right to left, $\mathfrak{B}_2$ creates the pair, then $U(x)U(z)$, then $\bar{\mathfrak{C}}_1^\dagger$ merges them. The splitting therefore happens *before* the shock wave and the merging *after*: the pair is a fluctuation of the incoming projectile wave function. This is spacelike, projectile DGLAP — the same object KLSZ resum — and not final-state (timelike) evolution. Splittings occurring after the shock wave, if present elsewhere in the $O(g^4)$ calculation, are a separate matter and belong to fragmentation.

3. Lemma 1: $\mathfrak{C}_1$ is the DGLAP $g \rightarrow gg$ vertex VERIFIED

Strip the colour and longitudinal prefactors from $\mathfrak{C}_1$ and write the tensor structure with $\zeta = p^+/k^+$, so that $k^+/p^+ = 1/\zeta$ and $k^+/(k^+ - p^+) = 1/(1 - \zeta)$:

$$V_{jk;i}(x) = x^i \delta_{jk} - \frac{1}{\zeta} x^j \delta_{ik} - \frac{1}{1 - \zeta} x^k \delta_{ij},$$

which is the standard light-cone-perturbation-theory $g \rightarrow gg$ vertex with x conjugate to the relative transverse momentum. Contract two of them over the daughter indices in $d = 2$, angular-average $x^m x^{m'} \rightarrow \frac{1}{2} x^2 \delta^{mm'}$ (the x^m/x^2 in $\mathfrak{C}_1$ supplies $1/x^2$ after the average), and restore the prefactor $[\sqrt{p^+(k^+ - p^+)}/k^+]^2 = \zeta(1 - \zeta)$:

$$\zeta(1-\zeta) \cdot \frac{1}{2} \sum_{j,k,m} V_{jk;i} V_{jk;i'} = \left[\frac{\zeta}{1-\zeta} + \frac{1-\zeta}{\zeta} + \zeta(1-\zeta) \right] \delta_{ii'} = \frac{P_{gg}(\zeta)}{2N_c} \delta_{ii'}.$$

The factor 2 from the tensor contraction and the $\frac{1}{2}$ from the angular average cancel exactly, which is why the result is the splitting function with no stray rational factor. The colour contraction $f^{abc} f^{a'bc} = N_c \delta^{aa'}$ supplies the N_c .

This is *identically* KLSZ's combination:

$$\frac{\xi^2 + (1-\xi)^2 + \xi^2(1-\xi)^2}{\xi(1-\xi)} = \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi) = \frac{P_{gg}(\xi)}{2N_c},$$

and with both endpoint poles plus-prescribed — KLSZ Eq. (27) — the integrand collapses to a polynomial, since $\frac{1}{\xi} + \frac{1}{1-\xi} = \frac{1}{\xi(1-\xi)}$ and $\xi^2 + (1-\xi)^2 - 1 = -2\xi(1-\xi)$:

$$\int_0^1 \frac{d\xi}{\xi_+(1-\xi)_+} (\xi^2 + (1-\xi)^2 + \xi^2(1-\xi)^2) = \int_0^1 d\xi (-2 + \xi - \xi^2) = -\frac{11}{6} = -\frac{\beta_0^g}{2N_c}, \quad \beta_0^g = \frac{11N_c}{3}.$$

So β_0^g is not put in by hand anywhere: it is the ζ -integral of the $g \rightarrow gg$ splitting function with the two soft poles removed. In the production cross section that ζ -integral is nothing other than the loop integral $\int dp^+ / 2\pi = (k^+ / 2\pi) \int_0^1 d\zeta$ already present in the first expression.

4. Lemma 2: the merging factorization VERIFIED

This is the technical heart of the identification, and it holds exactly, not just to logarithmic accuracy.

Work in the natural variables. The $\delta^{(2)}$ in $\bar{\mathfrak{C}}_1^\dagger(k^+, w - z; p^+, z - x)$ reads $\delta^{(2)}((w - z) + \zeta(z - x))$ and therefore states

$$w = (1 - \zeta)z + \zeta x,$$

i.e. the observed gluon sits at the momentum-fraction-weighted centroid of the two daughters — exactly the kinematics of KLSZ Eq. (25), where the daughters sit at $z + (1 - \xi)Z$ and $z - \xi Z$ about the parent z . Introduce

$$Z \equiv x - z \quad (\text{pair separation}), \quad u \equiv y - w \quad (\text{source to centroid}),$$

so that $z = w - \zeta Z$, $x = w + (1 - \zeta)Z$, and $\partial(w, Z) / \partial(z, x) = 1$: the change of variables $\int_{x,z} \rightarrow \int_w \int_Z$ is free.

Expanding the supplied $\mathfrak{B}_2$ about $Z = 0$: the energy denominator tends to $[p^+(y - x)^2 + (k^+ - p^+)(y - z)^2] k^+ \rightarrow (k^+)^2 u^2$, and the three structures in the bracket each produce a single $1/|Z|$,

$$\left[\dots \right]_{\text{sing}} = -\frac{u^n Z^m}{Z^2} \left[\delta^{ij} \delta^{nm} - \frac{k^+}{k^+ - p^+} \delta^{jn} \delta^{im} - \frac{k^+}{p^+} \delta^{in} \delta^{jm} \right],$$

whose bracket is *term by term* the bracket of $\mathfrak{C}_1$ with parent index n , parent momentum k^+ , and the relative vector Z in place of x . Restoring prefactors, the product $[ig/(\sqrt{2}\pi\sqrt{k^+})] \times [-g\sqrt{p^+(k^+ - p^+)}/(2\pi\sqrt{2k^+k^+})]$ equals $-ig^2\sqrt{p^+(k^+ - p^+)}/(4\pi^2(k^+)^2)$, which is minus the prefactor of $\mathfrak{B}_2$ divided by $(k^+)^2$ — the $4\pi^2$, the $\sqrt{p^+(k^+ - p^+)}$ and the $(k^+)^2$ all match with nothing left over. Hence

$$\mathfrak{B}_{2jk}^{be}(k^+ - p^+, -z; p^+, -x) \Big|_{Z \rightarrow 0} = -\mathfrak{A}_n^{(1)}(k^+, -w) \mathfrak{C}_{1k j n}(k^+; p^+, -Z) + O(Z^0),$$

with w the centroid. There is no $1/Z^2$ term; the remainder is genuinely $O(Z^0)$ and produces no logarithm. The overall sign follows the f^{cba} versus f^{abc} ordering in the supplied definitions and should be fixed against them; the modulus is exact.

Physically: *emit one gluon of momentum k^+ from the source, then split it*. That is precisely the statement that the two-gluon amplitude, in the collinear region, is the Weizsäcker–Williams field times the DGLAP vertex — the coordinate-space content of KLSZ Eq. (25).

The same expansion at *large* Z gives $\mathfrak{B}_2 \sim |u|/Z^2$ VERIFIED, so that $\bar{\mathfrak{C}}_1^\dagger \mathfrak{B}_2 \sim |u|/Z^3$. This is KLSZ's statement below their Eq. (12) that the pair-size integration is cut off in the infrared by the distance to the nearest source. It is what makes $\bar{Q}^2$ in

Sec. 8 depend on X and Y rather than on Q alone.

5. The transverse logarithm

Three ingredients now combine. First, the colour algebra. With $\bar{\mathfrak{C}}_1^{\dagger dca} \propto f^{dca}$ and $\mathfrak{B}_2^{be} \propto f^{bea''} \rho^{a''}$, and using invariance of f under the adjoint Wilson line,

$$f^{dca} U^{db}(x) U^{ce}(z) \Big|_{x \rightarrow z} = f^{bea'} U^{aa'}(z), \quad f^{bea'} f^{bea''} = N_c \delta^{a'a''}$$

VERIFIED numerically for $SU(3)$

so the two-Wilson-line structure collapses onto $N_c \times (\text{one Wilson line})$ — the production analogue of $D^{ab}(z, z) = N_c S^{ab}(z)$. Second, Lemma 2 turns $\bar{\mathfrak{C}}_1^\dagger \mathfrak{B}_2$ into $\bar{\mathfrak{C}}_1^\dagger \mathfrak{C}_1 \mathfrak{A}^{(1)}$, and the outer $\mathfrak{A}^{(1)}$ is already there; so what multiplies the LO cross section is exactly $|\mathfrak{C}_1|^2$, i.e. Lemma 1. Third, the transverse integral. In $d = 2$,

$$\int_{|Z| > 1/\mu} \frac{d^2 Z}{Z^2} e^{iq \cdot Z} = 2\pi \int_{1/\mu}^\infty \frac{dZ}{Z} J_0(qZ) = 2\pi \left[\ln \frac{2\mu}{q} - \gamma_E \right] = \pi \ln \frac{\mu_{\overline{\text{MS}}}^2}{q^2}, \quad \mu_{\overline{\text{MS}}}^2 = 4\mu^2 e^{-2\gamma_E},$$

VERIFIED numerically to 10^{-5} relative accuracy. The scale that emerges is exactly KLSZ's, who define $\mu^2 = \mu_{\overline{\text{MS}}}^2 e^{2\gamma}/4$ — a useful check that the coordinate-space cutoff $|Z| > 1/\mu$ is the same convention.

Putting the three together, the $Z \rightarrow 0$ region of the first supplied expression is the LO cross section times a universal scalar,

$$\left. \frac{d^3 N}{d^3 k} \right|_{1\text{st}, Z \rightarrow 0} = \Sigma(\mu, \bar{Q}) \frac{d^3 N_{\text{LO}}}{d^3 k} + \text{finite}, \quad \Sigma = \frac{\alpha_s N_c}{2\pi^2} \int_0^1 \frac{d\xi f(\xi)}{\xi_+(1-\xi)_+} \int_{1/\mu}^{1/\bar{Q}} \frac{d^2 Z}{Z^2} = -\frac{\alpha_s \beta_0^g}{4\pi} \ln \frac{\mu^2}{\bar{Q}^2}.$$

This is KLSZ Eq. (31) with no adjustable constant. The point to appreciate is that the coefficient was *not* fitted: Lemmas 1 and 2 fix the vertex algebra completely, so once the $Z \rightarrow 0$ limit is taken there is nothing left to choose. Any $O(g^4)$ production calculation built from these $\mathfrak{A}$, $\mathfrak{B}_2$, $\mathfrak{C}_1$ must produce this coefficient.

6. The one thing that can break the identification: the rapidity subtraction

Everything above rests on the **double** plus prescription. If your scheme regulates only $\zeta \rightarrow 0$, the number is not $-11/6$ and the whole β_0^g identification fails. This is the single check to do first.

KLSZ subtract both poles because both are already generated by LO JIMWLK: the $\xi \rightarrow 0$ and $\xi \rightarrow 1$ regions of the splitting function are the eikonal emissions that the low- x evolution resums, so including them again in the collinear term would double count. In the production cross section the same statement reads: the $\zeta \rightarrow 0$ and $\zeta \rightarrow 1$ endpoints of $\int dp^+ / 2\pi$ are the rapidity divergences, and they are subtracted into the LO BK/JIMWLK evolution of the target Wilson-line correlators. The structure is visible in $\mathfrak{C}_1$ itself, which carries both $1/p^+$ and $1/(k^+ - p^+)$ explicitly.

Concretely, what must be true is

$$\int_0^1 d\zeta \left[\frac{f(\zeta)}{\zeta(1-\zeta)} - \underbrace{\frac{f(0)}{\zeta}}_{\zeta \rightarrow 0 \text{ rapidity subtraction}} - \underbrace{\frac{f(1)}{1-\zeta}}_{\zeta \rightarrow 1 \text{ rapidity subtraction}} \right], \quad f(\zeta) = \zeta^2 + (1-\zeta)^2 + \zeta^2(1-\zeta)^2,$$

with $f(0) = f(1) = 1$. Note that the two subtractions are *not* symmetric in physical origin even though they are in the algebra: $\zeta \rightarrow 0$ makes the daughter at x soft, $\zeta \rightarrow 1$ makes the daughter at z soft, and in a production calculation these may be handled by different pieces of the rapidity-subtraction machinery. Verify that after your subtraction the leftover ζ -integrand is the polynomial $-2 + \zeta - \zeta^2$. If it is, the coefficient is β_0^g and the rest of this note applies verbatim.

A consequence worth keeping in mind, and stressed by KLSZ in their discussion: because both poles are removed, the "real" term in the dressed-gluon definition enters with a *negative* coefficient. These are not probabilities and the dressed states are not the usual normalized dressed partons. Nothing is wrong; but signs that look pathological are expected, and the resummed Wilson line will come out *less* saturated, not more (Sec. 9).

7. The dressed Wilson line for production, and the split of the $2\beta_0$

The second supplied expression, $\mathfrak{A}^{\dagger(1)} \otimes \mathfrak{A}^{(3)}$, is the one-Wilson-line structure. Its $\ln \mu^2$ content should be the production analogue of KLSZ Eq. (7): a coefficient of $\ln \mu_{\overline{\text{MS}}}^2$ equal to $2\beta_0$ relative to LO — twice what the running coupling can absorb. Half of it is the gluon wave-function renormalization sitting inside $\mathfrak{A}^{(3)}$,

$$Z^{1/2}(Q^2) = 1 + \frac{\alpha_s}{8\pi} \beta_0 \ln \frac{Q^2}{\mu^2},$$

and it is this half, and only this half, that belongs to the running of α_s . The other half is Σ of Sec. 5 — DGLAP.

The clean way to organize this, following KLSZ, is to re-express the cross section in terms of a gluon dressed to resolution $\bar{Q}$. In production the dressing must be applied to the amplitude and the conjugate *separately*, because $w \neq w'$:

$$U_{\bar{Q}}^{ab}(w) = \left[1 + \frac{\alpha_s \beta_0^g}{4\pi} \ln \frac{\mu^2}{\bar{Q}^2} \right] U^{ab}(w) - \frac{\alpha_s \beta_0^g}{4\pi^2 N_c} \int_{|Z| < 1/\bar{Q}} \frac{d^2 Z}{Z^2} \text{Tr} \left[T^a U(w + \frac{Z}{2}) T^b U^\dagger(w - \frac{Z}{2}) \right].$$

Substituting $U \rightarrow U_{\bar{Q}}$ in the LO cross section then does three things at once, and this is the whole mechanism:

1. the term linear in U supplies $-\frac{\alpha_s \beta_0^g}{4\pi} \ln(\mu^2/\bar{Q}^2)$, which combines with half of the $2\beta_0$ log of $\mathfrak{A}^{(3)}$ to leave exactly the UV logarithm that the running coupling absorbs;
2. the bilinear term cancels the $Z \rightarrow 0$ divergence of the first supplied expression — this is Σ , and it is the cancellation that makes the two-Wilson-line term UV finite;
3. what is left over is a *finite* $\ln(X^2 \bar{Q}^2)$, which is the DGLAP logarithm and is the object to be resummed.

For the running-coupling half, production is actually cleaner than the Hamiltonian. KLSZ must choose how to distribute $\alpha_s(X^{-2})$ and $\alpha_s(Y^{-2})$ over the kernel (their footnote 8, the parameter λ ; $\lambda = \frac{1}{2}$ is their choice, $\lambda = 1$ the "triumvirate"). Here the two WW fields sit in separate factors — one in the amplitude, one in the conjugate — so the natural prescription is simply

$$g \longrightarrow g(1/(y-w)^2) \text{ inside } \mathfrak{A}^{(1)}(k^+, -w), \quad g \longrightarrow g(1/(y'-w')^2) \text{ inside } \mathfrak{A}^{\dagger(1)}(k^+, -w'),$$

i.e. set the coupling at the distance between the emitter and the emission point, separately in amplitude and conjugate. This reproduces KLSZ's $\lambda = \frac{1}{2}$ combination $g(X^{-2})g(Y^{-2})$ on squaring, and it is local in w , so it is trivial to implement numerically.

8. The resolution scale in production: what actually cuts off $|Z|$

This is where production genuinely differs from the Hamiltonian, and it is worth being careful because the naive guess is wrong.

The tempting argument is that since we measure k , the phase $e^{-ik(w'-w)}$ must oscillate in Z and cut the pair-size integral at $|Z| \sim 1/k_T$. It does not. Using the centroid variables of Sec. 4, $\int_{x,z} = \int_w \int_Z$ with unit Jacobian, and the phase depends on w alone:

$$e^{-ik(w'-w)} \text{ is independent of } Z.$$

The Z -dependence of the integrand sits entirely in $U(w + (1-\zeta)Z)U(w - \zeta Z)$ and in the source-distance dependence of $\mathfrak{B}_2$. Hence the two cutoffs are the same two that KLSZ identify:

- **Target decorrelation.** Two eikonal factors separated by $|Z| \gg 1/Q_T$ fluctuate independently and their product averages to zero in the target ensemble. This caps the logarithmic region at $|Z| \sim 1/Q_T$.
- **Distance to the emitter.** $\mathfrak{B}_2 \sim |u|/Z^2$ at large Z (Sec. 4), so a pair larger than the emitter–gluon distance is suppressed. This caps $|Z|$ at $|X|$ (respectively $|Y|$ in the conjugate).

The resolution scale in production is therefore KLSZ's, with X, Y read off the production kinematics as in Sec. 1:

$$\bar{Q}^2 = \max \left\{ Q_T^2, \frac{1}{X^2}, \frac{1}{Y^2} \right\}, \quad \tilde{Q}^2 = \max \left\{ \frac{1}{X^2}, \frac{1}{Y^2} \right\}, \quad X = y - w, \quad Y = y' - w'.$$

Now k_T enters, but indirectly. The Fourier transform at fixed k weights the Weizsäcker–Williams field so that $\int d^2 X \frac{X^i}{X^2} e^{ik \cdot X} \propto k^i / k^2$ is dominated by $|X| \lesssim 1/k_T$. So $1/X^2 \sim k_T^2$, and

$$\ln(X^2 \bar{Q}^2) \simeq \begin{cases} \ln \frac{Q_T^2}{k_T^2}, & k_T \lesssim Q_T \quad (\text{large}), \\ 0, & k_T \gtrsim Q_T \quad (\text{no DGLAP log}). \end{cases}$$

The physical conclusion. The DGLAP-like logarithm in single-inclusive gluon production is large in the *saturation region* $k_T \lesssim Q_s$, and switches off above it. At large k_T the emission point is close to the emitter, the pair cannot be resolved, and $\bar{Q}^2 \rightarrow 1/X^2$ makes the log vanish — there, all that survives is the running coupling at the natural scale $\alpha_s(k_T^2)$. This is the exact production counterpart of KLSZ's condition $Q_T \gg Q_P$: the projectile's colour correlation length must exceed the target's, and in production the projectile scale probed is $Q_P \sim k_T$.

9. The resummed production cross section

The resummation condition is that the cross section not depend on the arbitrary scale $\bar{Q}$. The dressed Wilson line of Sec. 7, written differentially, obeys the DGLAP-like equation (KLSZ Eq. 43, transposed to w):

$$\frac{\partial}{\partial \ln \bar{Q}^2} U_{\bar{Q}}^{ab}(w) = -\frac{\alpha_s \beta_0^g}{4\pi} \left[U_{\bar{Q}}^{ab}(w) - \frac{1}{N_c} \int \frac{d\phi}{2\pi} D_{\bar{Q}}^{ab} \left(w + \frac{e_\phi}{2\bar{Q}}, w - \frac{e_\phi}{2\bar{Q}} \right) \right], \quad D_{\bar{Q}}^{ab}(z_1, z_2) = \text{Tr} [T^a U_{\bar{Q}}(z_1) T^b U_{\bar{Q}}^\dagger(z_2)],$$

to be integrated from the initial condition $U_{Q_T}(w) = U(w)$ — above Q_T the cascade fits inside one target correlation length and is unresolved — down to $\bar{Q} = \bar{Q}(w)$, where no large logarithm is left. Both limits carry over unchanged from KLSZ, because both were arguments about the target correlation length and the emitter–gluon distance, neither of which the measurement modifies.

Saturated target

When Q_T is a saturation momentum, Wilson lines separated by more than $1/Q_T$ have vanishing correlation, so the bilinear term drops and the equation linearizes to $\partial_{\ln \bar{Q}^2} U_{\bar{Q}} = -\frac{\alpha_s \beta_0^g}{4\pi} U_{\bar{Q}}$, with the power-law solution of KLSZ Eq. (65). The resummed production cross section is then LO with running coupling and one local, multiplicative modification of each Wilson line:

$$\left. \frac{d^3 N}{d^3 k} \right|_{\text{resummed}} = \frac{1}{(2\pi)^3} \int_{w, w'} e^{-ik(w'-w)} \left[-\bar{\mathfrak{A}}_i^{\dagger(1)a}(w') + \mathfrak{A}_i^{\dagger(1)b'}(w') \mathcal{K}(w') U^{b'a}(w') \right] \left[\mathcal{K}(w) U^{ab}(w) \mathfrak{A}_i^{(1)b}(w) - \bar{\mathfrak{A}}_i^{(1)a}(w) \right]$$

$$\mathcal{K}(w) = \theta(\bar{Q}(w) - Q_T) + \theta(Q_T - \bar{Q}(w)) \left[\frac{Q_T^2}{\bar{Q}^2(w)} \right]^\gamma, \quad \gamma \equiv \frac{\alpha_s \beta_0^g}{4\pi} = \frac{11 N_c \alpha_s}{12\pi},$$

together with $g \rightarrow g(1/X^2)$ inside $\mathfrak{A}^{(1)}$ and $g \rightarrow g(1/Y^2)$ inside $\mathfrak{A}^{\dagger(1)}$ (Sec. 7). $\bar{Q}^2(w) = \max\{1/X^2, 1/Y^2\}$.

Two features of this deserve comment. First, $\mathcal{K}$ multiplies only the *interacting* piece of the amplitude; the inhomogeneous $\bar{\mathfrak{A}}$ term is untouched, so the correction does not simply rescale the cross section — it changes the balance between the two terms, and hence the shape of the k_T spectrum, not just its normalization. Second, $\mathcal{K} > 1$ for $k_T < Q_T$. Following KLSZ's sign discussion (Sec. 6), this means the dressed target looks *less* saturated: the effective saturation momentum probed in production is reduced, and the expected net effect is a suppression of gluon production at $k_T \ll Q_s$. The precise spectrum requires doing the convolution; the statement that survives without it is that the modification is a power of Q_s^2/k_T^2 with anomalous dimension γ .

Dilute target

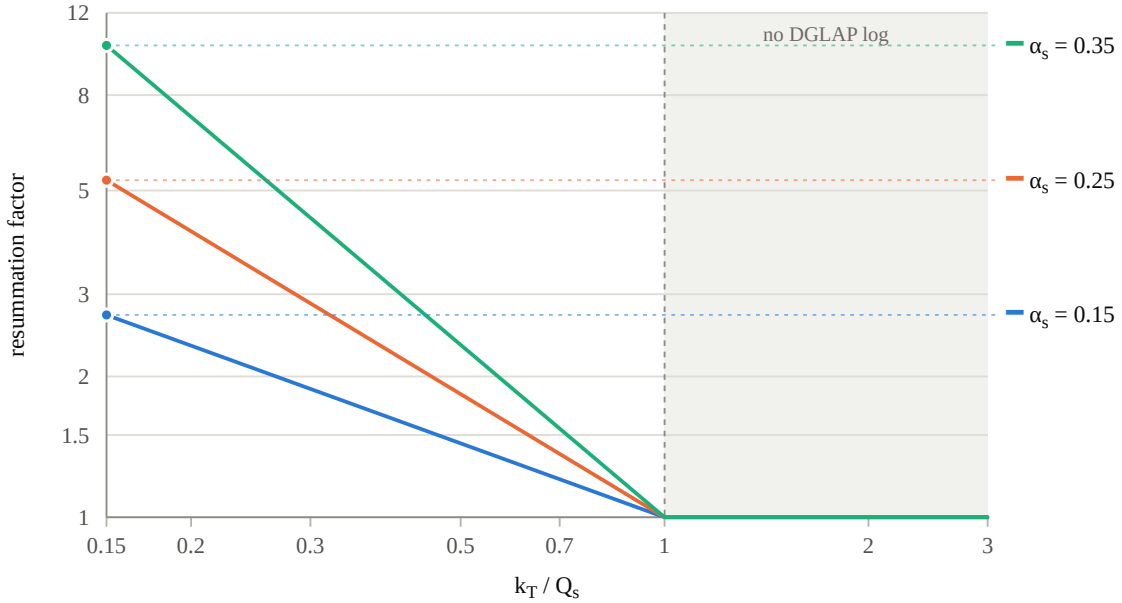
If neither projectile nor target is saturated, write $U = 1 + iT^a \alpha^a$ and linearize. Then the real term cannot be dropped and one gets KLSZ Eq. (47), whose solution is the smearing of the emission point over a power-law kernel:

$$\alpha_{\tilde{Q}}(w) = \int d^2y G_{\tilde{Q}}(w-y) \alpha(y), \quad G_{\tilde{Q}}(x) \simeq \left(\frac{Q_s^2}{\tilde{Q}^2}\right)^\gamma \delta^{(2)}(x) + \frac{\gamma \tilde{Q}^2}{\pi} \left[1 - \left(\frac{1}{\tilde{Q}^2 x^2}\right)^{1+\gamma}\right] \text{ for } \frac{1}{\tilde{Q}_T} < |x| < \frac{1}{\tilde{Q}},$$

falling as $(\tilde{Q}|x|)^{-5/2}$ beyond, and normalized by $\int d^2x G_{\tilde{Q}}(x) = 1$. In production this means the Lipatov vertex is evaluated not at the emission point but smeared over a transverse region of size $1/\tilde{Q} \sim 1/k_T$ with a slowly decreasing power-law weight. Because the weight falls so slowly, the smearing term contributes at $O(1)$ even though it carries an explicit α_s — it must be kept.

How big is it?

In the saturated regime the correction to the interacting part of the cross section is $\mathcal{K}(w)\mathcal{K}(w') \simeq (Q_s^2/k_T^2)^{2\gamma}$. With $\gamma = 11N_c\alpha_s/12\pi$ and $N_c = 3$, $\gamma \simeq 0.875\alpha_s$ — not a small anomalous dimension.



Resummation factor $(Q_s^2/k_T^2)^{2\gamma}$ multiplying the interacting part of the production amplitude squared, versus k_T/Q_s . Log-log axes, so each curve is a straight line of slope $-2\gamma = -\alpha_s\beta_0^g/2\pi$. The logarithm switches off above Q_s . Fixed coupling; running α_s in the DGLAP kernel (KLSZ Eq. 57) tames the small- k_T growth and replaces the power by a ratio of logarithms.

k_T/Q_s	$\alpha_s = 0.15$	$\alpha_s = 0.25$	$\alpha_s = 0.35$
0.15	2.71	5.26	10.23
0.20	2.33	4.09	7.19
0.30	1.88	2.87	4.37
0.50	1.44	1.83	2.34
0.70	1.21	1.37	1.55
1.00	1.00	1.00	1.00

Even at $\alpha_s = 0.15$ the factor reaches ~ 2 by $k_T \simeq Q_s/3$. Whatever the eventual sign of the effect on the spectrum, a correction of this size is not something an NLO calculation can leave unresummed in the region where saturation physics is supposed to be predictive. That is the concrete sense in which the answer to the original question is yes.

10. What is not settled, and what to compute next

The identification of Secs. 3–5 is, I believe, airtight given the supplied vertices. What follows is not, and each item is genuine work.

Item	Status	What to do
Rapidity subtraction gives the double plus prescription	assumed	The decisive check (Sec. 6). Confirm the leftover ζ -integrand is $-2 + \zeta - \zeta^2$. If only $\zeta \rightarrow 0$ is subtracted, the coefficient is not β_0^g and the mapping fails.
$\ln \mu^2$ coefficient of $\mathfrak{A}^{(3)}$ is $2\beta_0$	expected, not verified	$\mathfrak{A}^{(3)}$ was not supplied. Extract its UV log; it should be $\beta_0 [\frac{1}{2} \ln X^2 \mu^2 + \frac{1}{2} \ln Y^2 \mu^2]$ relative to LO, i.e. $2\beta_0$, matching KLSZ Eq. (7). If the coefficient comes out β_0 rather than $2\beta_0$, then $\mathfrak{A}^{(3)}$ already has $Z^{1/2}$ applied and the split in Sec. 7 must be redone.
UV cancellation between the two expressions	structurally clear	Verify that Σ of Sec. 5 cancels the $\mathfrak{A}^{(3)}$ divergence <i>separately in amplitude and conjugate</i> . Because $w \neq w'$ this is two cancellations, not one; the factor 2 in front of the first supplied expression is presumably the mirror term and should be checked to be exactly that.
Overall sign in Lemma 2	convention	Fix f^{cba} versus f^{abc} in the supplied $\mathfrak{B}_2$ and $\mathfrak{C}_1$. The modulus of the factorization is exact; only the sign is convention-dependent.
N_f and the full β_0	missing	The supplied terms are pure glue and give $\beta_0^g = 11N_c/3$. The full $\beta_0 = (11N_c - 2N_f)/3$ needs $g \rightarrow q\bar{q}$: a $\mathfrak{C}_1$ -like vertex with quark lines and the corresponding $\mathfrak{B}_2$, giving KLSZ's $K_{q\bar{q}}$ and $\beta_0^q = -2N_f/3$. Note KLSZ's warning that $\mathcal{V}_Q^\dagger \mathcal{V}_Q \neq 1$ for dressed quarks, so the dressed quark Wilson line satisfies an independent evolution equation and cannot be inferred from the gluon one.
Other $O(g^4)$ two- ρ structures	unchecked	In particular $\int dp^+ \mathfrak{B}_2 ^2$ with the unobserved gluon integrated. This is a genuine real correction and produces collinear logs when the unobserved gluon becomes collinear to the observed one. Those are <i>final-state</i> logs: they do not cancel against the virtual ones here because measuring k breaks the inclusive sum, and they belong to fragmentation, not to the projectile DGLAP discussed above. They must be separated before the β_0^g extraction is claimed to be complete.
Post-shock splittings	unchecked	If the $O(g^4)$ calculation contains a single U followed by $\mathfrak{C}_1$ (splitting <i>after</i> the target), that is timelike evolution and is a separate resummation. Do not merge it with Secs. 5–9.
Sign of the net effect on d^3N/d^3k	open	Sec. 9 gives the substitution rule; the spectrum requires the convolution with a model target. Worth doing numerically — the substitution is local in w , so it is cheap.

Suggested order of work

1. Check the rapidity subtraction leaves $-2 + \zeta - \zeta^2$ (Sec. 6). Everything depends on this and it is a half-page calculation.
2. Extract the $\ln \mu^2$ coefficient of $\mathfrak{A}^{(3)}$ and confirm $2\beta_0$.
3. Do the $Z \rightarrow 0$ subtraction of the first expression explicitly using Lemma 2 and confirm the two divergences cancel, amplitude and conjugate separately.
4. Substitute $U \rightarrow U_{\bar{Q}}$ and read off the residual $\ln(X^2 \bar{Q}^2)$; confirm $\bar{Q}^2 = \max\{Q_T^2, 1/X^2, 1/Y^2\}$ by tracking which cutoff wins.
5. Only then add the $q\bar{q}$ channel and do the numerics.

11. Verification log

Run `python3 verify.py` in this directory. Requires `sympy`, `numpy`, `scipy`.

Check	Method	Result
$\mathfrak{B}_2$ has no $1/Z^2$; its $1/ Z $ residue equals $(u^n/u^2)(k^+)^{-2} \times \mathfrak{C}_1$ tensor, all four (i, j)	SymPy, exact limit in $d = 2$	pass
Prefactors: $\mathfrak{A}^{(1)} \times \mathfrak{C}_1 = -\mathfrak{B}_2/(k^+)^2$ (incl. the $4\pi^2$)	SymPy	pass, difference identically 0
$\mathfrak{B}_2 \sim 1/Z^2$ at large Z	SymPy asymptotics	pass
$\zeta(1 - \zeta) \cdot \frac{1}{2} \sum VV = P_{gg}(\zeta)/2N_c$	SymPy tensor contraction	pass, difference identically 0
$[\xi^2 + (1 - \xi)^2 + \xi^2(1 - \xi)^2]/\xi(1 - \xi) = P_{gg}/2N_c$	SymPy	pass
Double-plus integral $= -11/6 = -\beta_0^g/2N_c$	SymPy, integrand $-2 + \xi - \xi^2$	pass
$f^{dca}U^{db}U^{ce} = f^{bea'}U^{aa'}; f^{bea'}f^{bea''} = N_c\delta$	numerical $SU(3)$, random group element	pass, $N_c = 3.000$
$\int_{ Z >1/\mu} \frac{d^2Z}{Z^2} e^{iq \cdot Z} = \pi \ln(\mu_{\overline{\text{MS}}}^2/q^2), \mu_{\overline{\text{MS}}}^2 = 4\mu^2 e^{-2\gamma_E}$	numerical quadrature, Bessel	pass to $\sim 10^{-5}$ rel.

Reference: A. Kovner, M. Lublinsky, V. V. Skokov, Z. Zhao, *Not all that is β_0 is β -function: the DGLAP resummation and the running coupling in NLO JIMWLK*, arXiv:2308.15545 [hep-ph]. Equation numbers cited in this note refer to that paper.